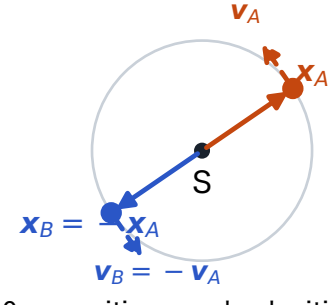
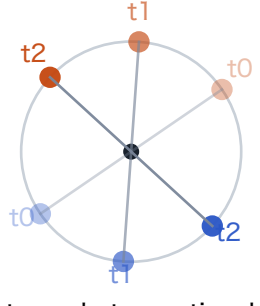
Fig. 1 The closed pair: one closure fixes position and momentum at once; a readout reads one end

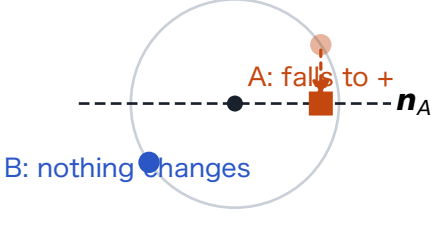
(a) antipodal pair: $\boldsymbol{x}_B = -\boldsymbol{x}_A$, $\boldsymbol{p}_B = -\boldsymbol{p}_A$



(b) the closure is kept as time runs



(c) reading A is no event at B



closure $q_A + q_B = 0 \Rightarrow$ positions and velocities sum to zero at each t; both turn at the same rate and stay antipodal ($q_B = -q_A$ at all times)

A is read destructively against its axis (P4, known); B is untouched